

Task 3: AHB System with 4 Slave

Objective

The objective of this task is to develop a comprehensive architecture of an AHB (Advanced High-performance Bus) system containing one master and four slaves. The system demonstrates how the AHB bus connects the master to multiple slave devices using address decoding and multiplexing logic.

System Architecture

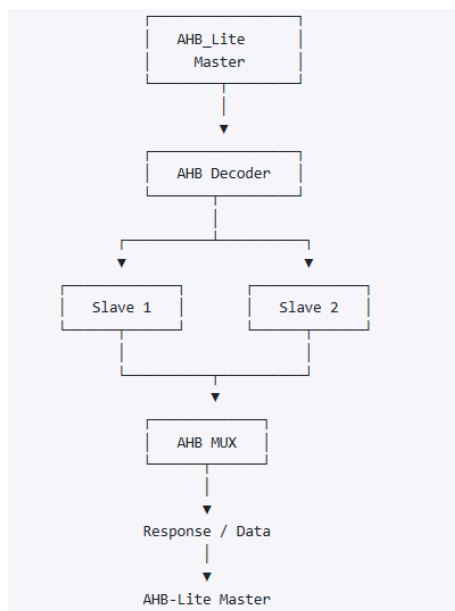
The AHB system consists of the following major components:

- AHB Master
- AHB Bus Interconnect
- Address Decoder
- Four AHB Slaves
- Read Data Multiplexer
- Response Multiplexer

The AHB Master initiates read and write transactions by generating the required address, control, and data signals. The address decoder examines the address generated by the master and selects the appropriate slave.

The four slaves can represent different memory-mapped peripherals such as RAM, ROM, UART, GPIO, or other system peripherals.

Block Diagram



Signal Interconnections

The main AHB signals exchanged between the master, bus interconnect, and slaves include:

- HADDR – Carries the address of the current transfer.
- HTRANS – Indicates the type of transfer such as IDLE, BUSY, NONSEQ, or SEQ.
- HWRITE – Indicates whether the operation is a read or write.
- HSIZE – Specifies the size of the transfer.
- HBURST – Defines the burst type.
- HPROT – Provides additional protection/control information.
- HWDATA – Carries data from the master to the selected slave.
- HRDATA – Carries read data from the selected slave to the master.
- HREADY – Indicates whether the transfer has completed.
- HRESP – Indicates the response status of the transfer, such as OKAY or ERROR.
- HSEL – Slave select signal generated by the address decoder.

Address Decoding

The address decoder determines which slave should respond to a transaction based on the value of HADDR.

An example memory map is:

Address Range	Selected Slave	Example Device
0x0000 – 0x0FFF	HSEL0	RAM
0x1000 – 0x1FFF	HSEL1	ROM
0x2000 – 0x2FFF	HSEL2	UART
0x3000 – 0x3FFF	HSEL3	GPIO

Operational Flow

The operation of the AHB system can be summarized as follows:

1. The AHB master initiates a transaction by placing an address on HADDR.
2. The master generates the required control signals such as HTRANS, HWRITE, HSIZE, and HBURST.
3. The address decoder examines HADDR.
4. The decoder activates the HSEL signal corresponding to the selected slave.
5. For a write operation, HWDATA is transferred from the master to the selected slave.
6. For a read operation, the selected slave places data on HRDATA.
7. The selected slave controls HREADY to indicate whether the transfer is complete or requires additional wait states.

8. HRESP indicates whether the transaction completed successfully or resulted in an error.
9. The response and read data are routed back to the master through the corresponding multiplexing logic.
10. The master completes the transfer and proceeds to the next transaction.

Conclusion

An AHB system with multiple slaves uses a shared bus architecture together with address decoding to determine which slave responds to each transaction. The address decoder ensures that only the appropriate slave is selected, while multiplexing logic routes read data and response signals back to the master. This architecture allows multiple memory and peripheral devices to communicate efficiently with a single AHB master while maintaining a structured memory-mapped system.